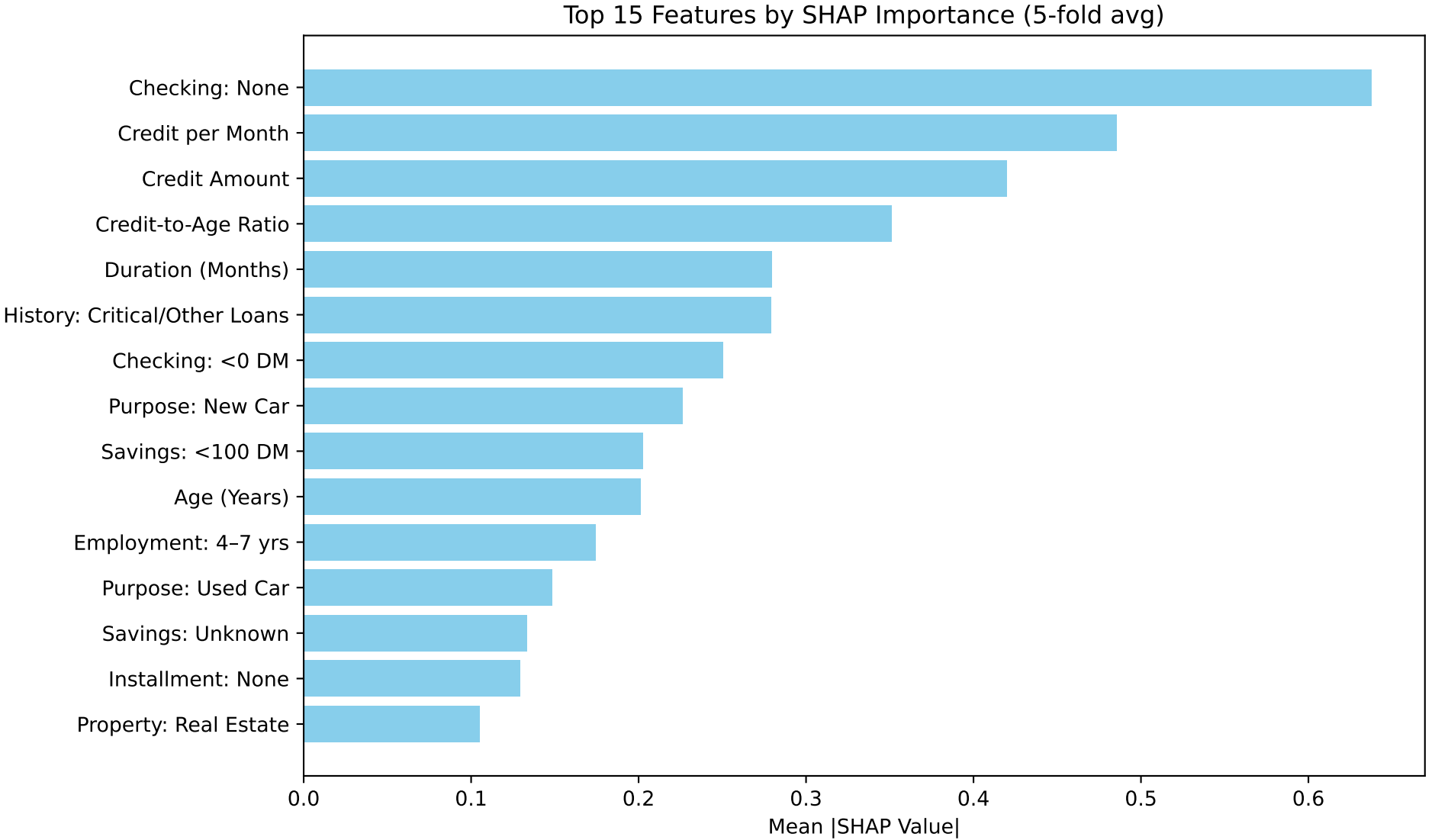
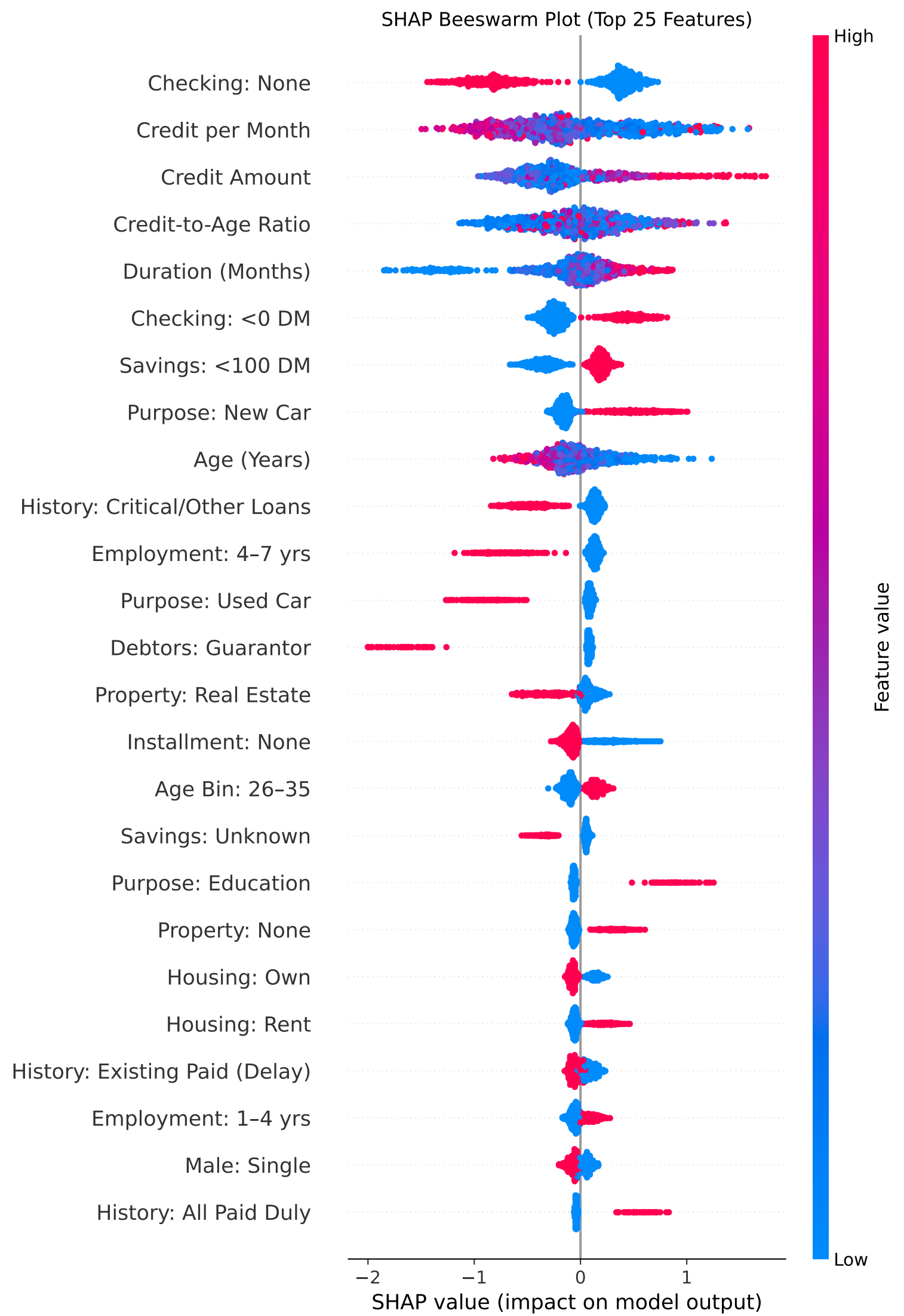
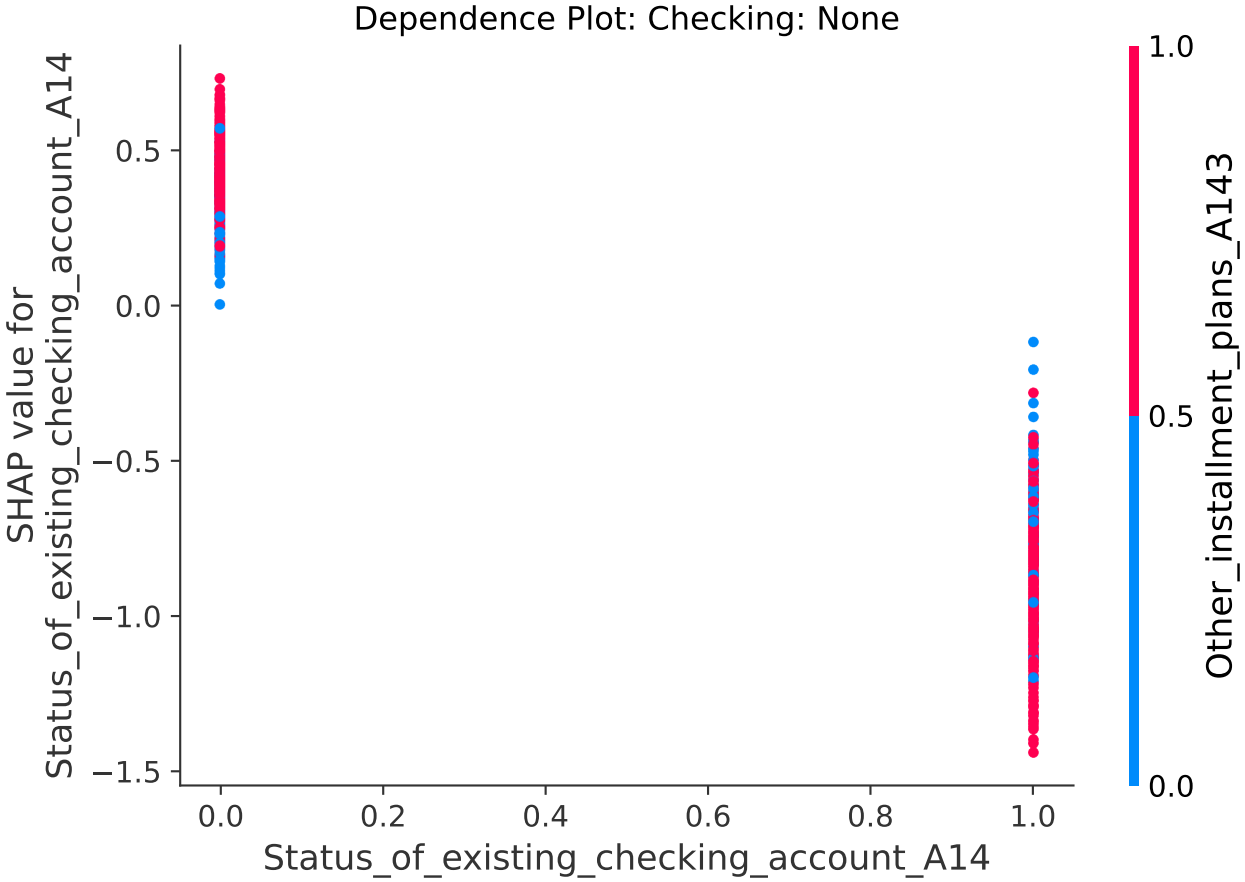
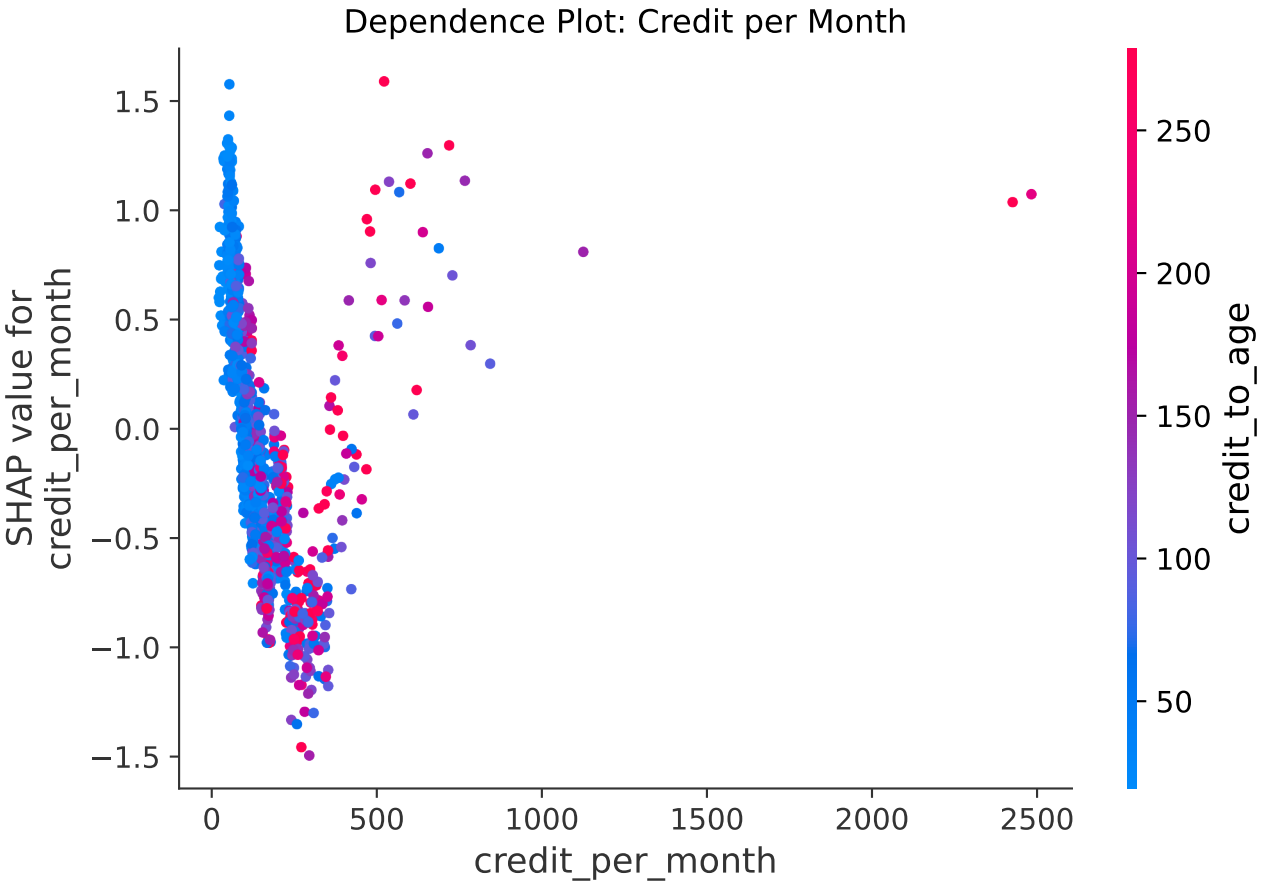
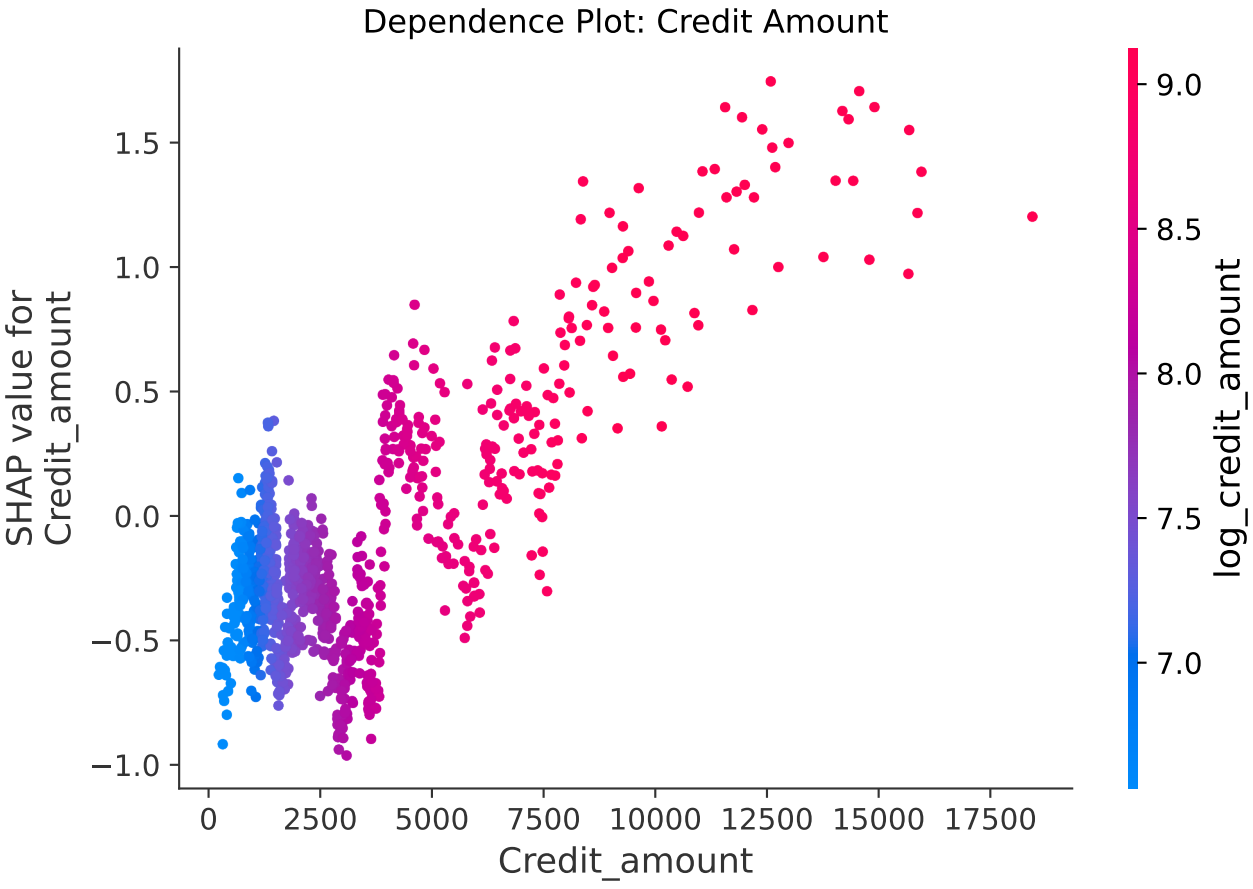
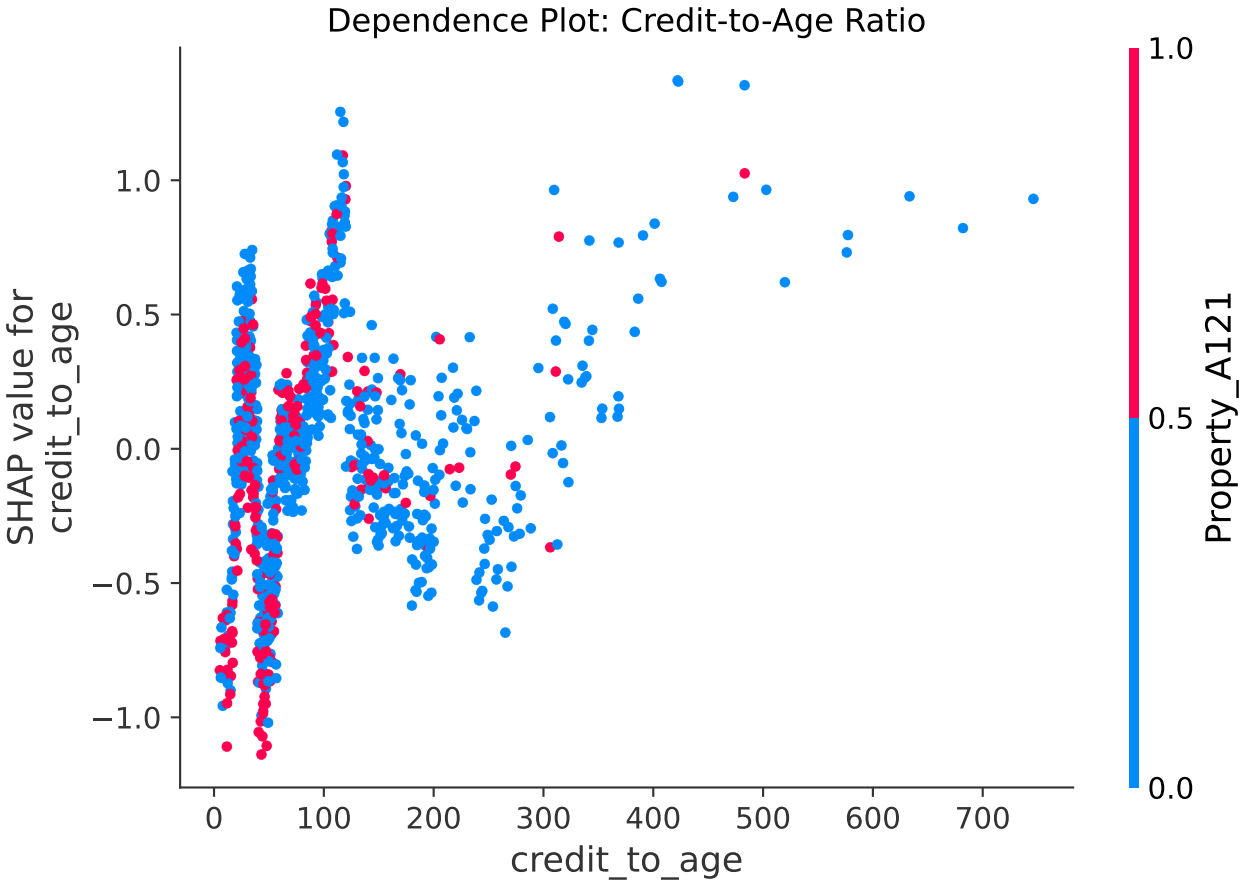

Top 15 Features by SHAP Importance (5-fold avg)
Checking: None
Credit per Month
Credit Amount
Credit-to-Age Ratio
Duration (Months)
History: Critical/Other Loans
Checking: <0 DM
Purpose: New Car
Savings: <100 DM
Age (Years)
Employment: 4–7 yrs
Purpose: Used Car
Savings: Unknown
Installment: None
Property: Real Estate
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Mean |SHAP Value|

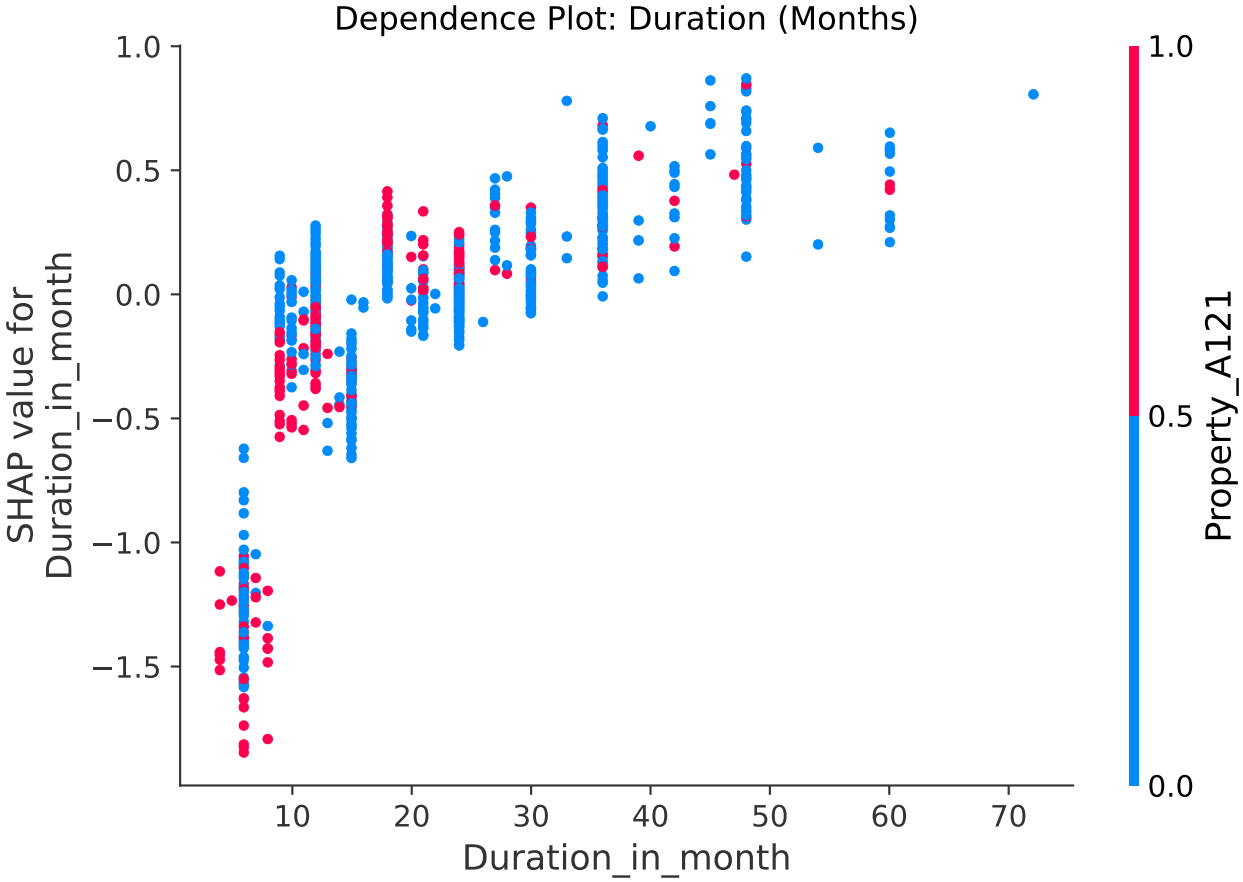

SHAP Beeswarm Plot (Top 25 Features)
Checking: None
Credit per Month
Credit Amount
Credit-to-Age Ratio
Duration (Months)
Checking: <0 DM
Savings: <100 DM
Purpose: New Car
Age (Years)
History: Critical/Other Loans
Employment: 4–7 yrs
Purpose: Used Car
Debtors: Guarantor
Property: Real Estate
Installment: None
Age Bin: 26–35
Savings: Unknown
Purpose: Education
Property: None
Housing: Own
Housing: Rent
History: Existing Paid (Delay)
Employment: 1–4 yrs
Male: Single
History: All Paid Duly
−2
−1
0
1
SHAP value (impact on model output)
High
Low
Feature value

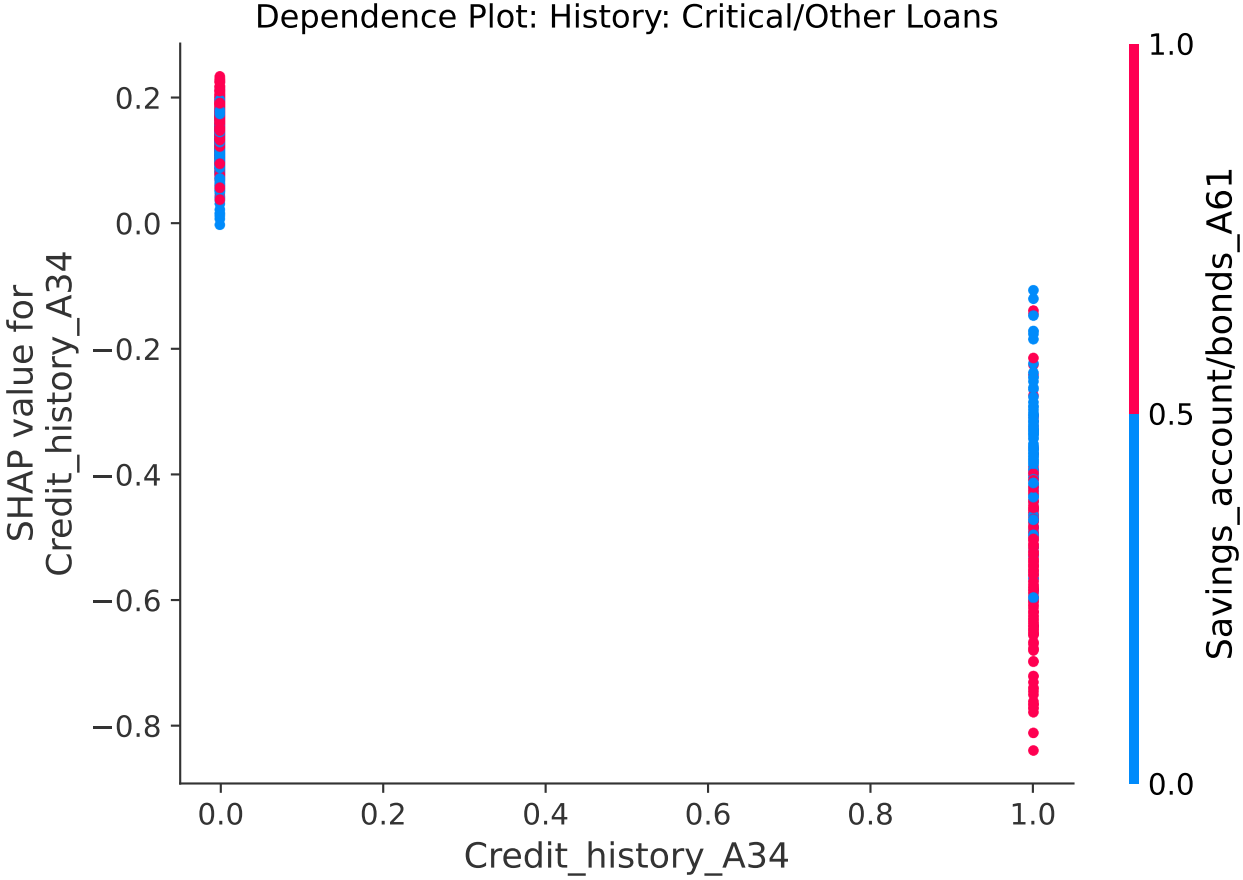

Dependence Plot: Checking: None
SHAP value for
Status_of_existing_checking_account_A14
0.5
0.0
−0.5
−1.0
−1.5
0.0
0.2
0.4
0.6
0.8
1.0
Status_of_existing_checking_account_A14
1.0
0.5
0.0
Other_installment_plans_A143

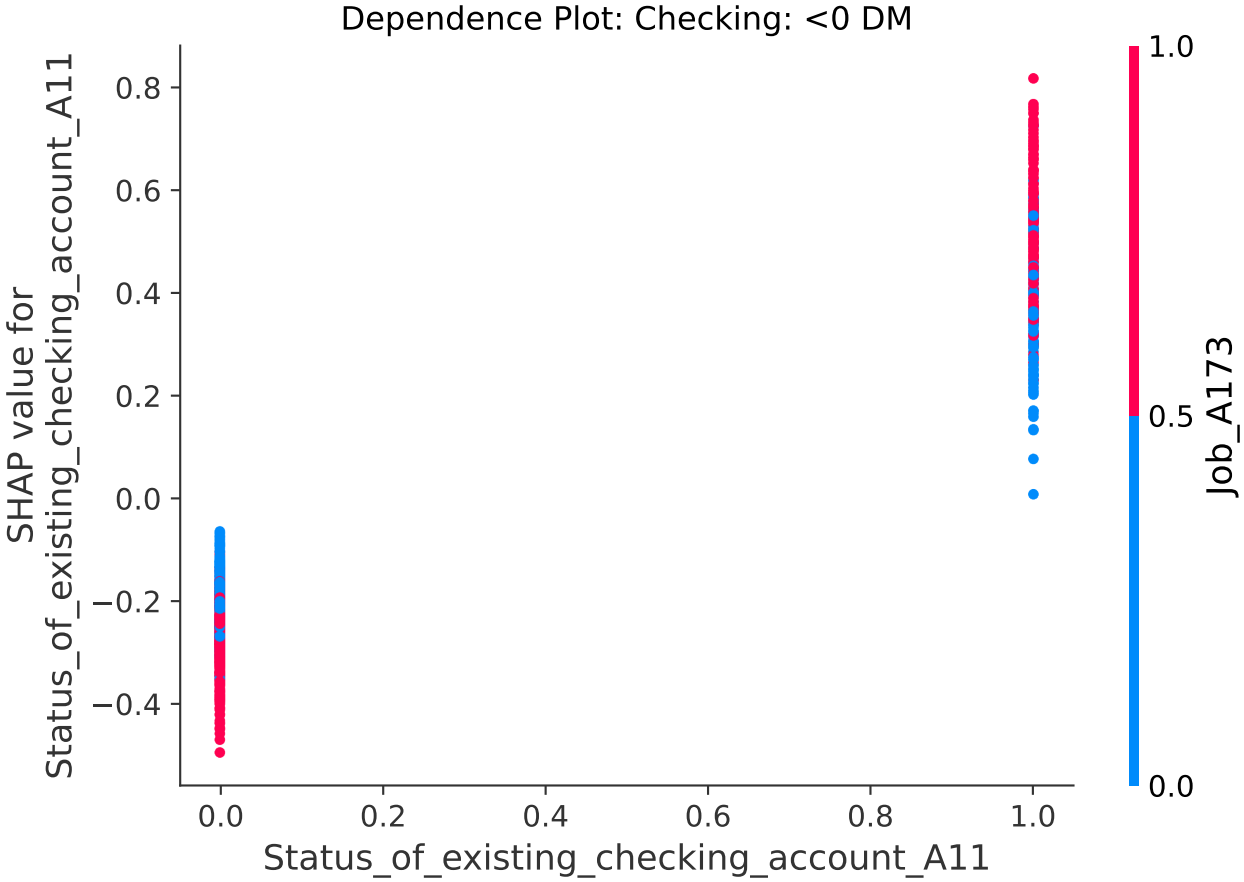

Dependence Plot: Credit per Month
SHAP value for
credit_per_month
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
2000
2500
credit_per_month
250
200
150
100
50
credit_to_age

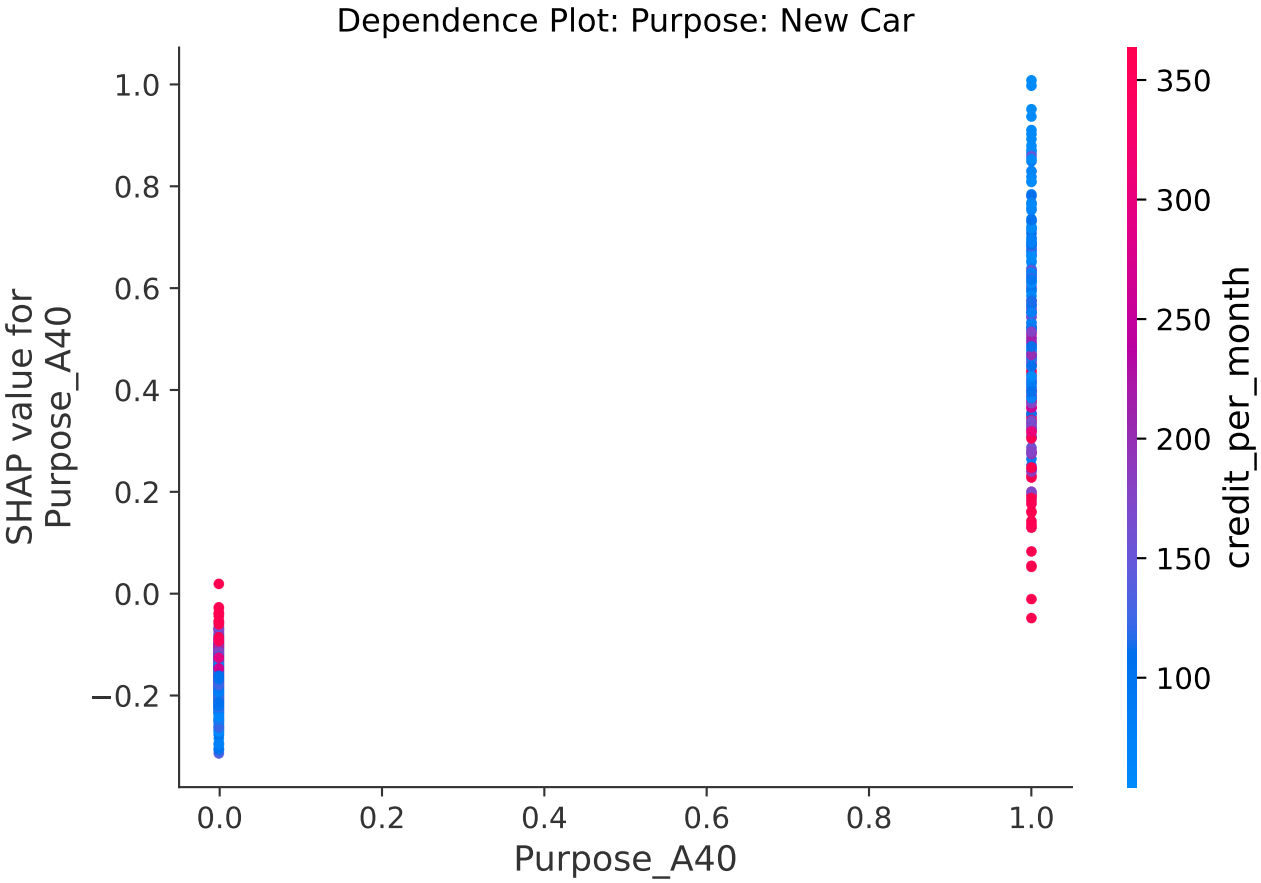

Dependence Plot: Credit Amount
SHAP value for Credit_amount
1.5
1.0
0.5
0.0
−0.5
−1.0
0
2500
5000
7500
10000
12500
15000
17500
Credit_amount
log_credit_amount
9.0
8.5
8.0
7.5
7.0

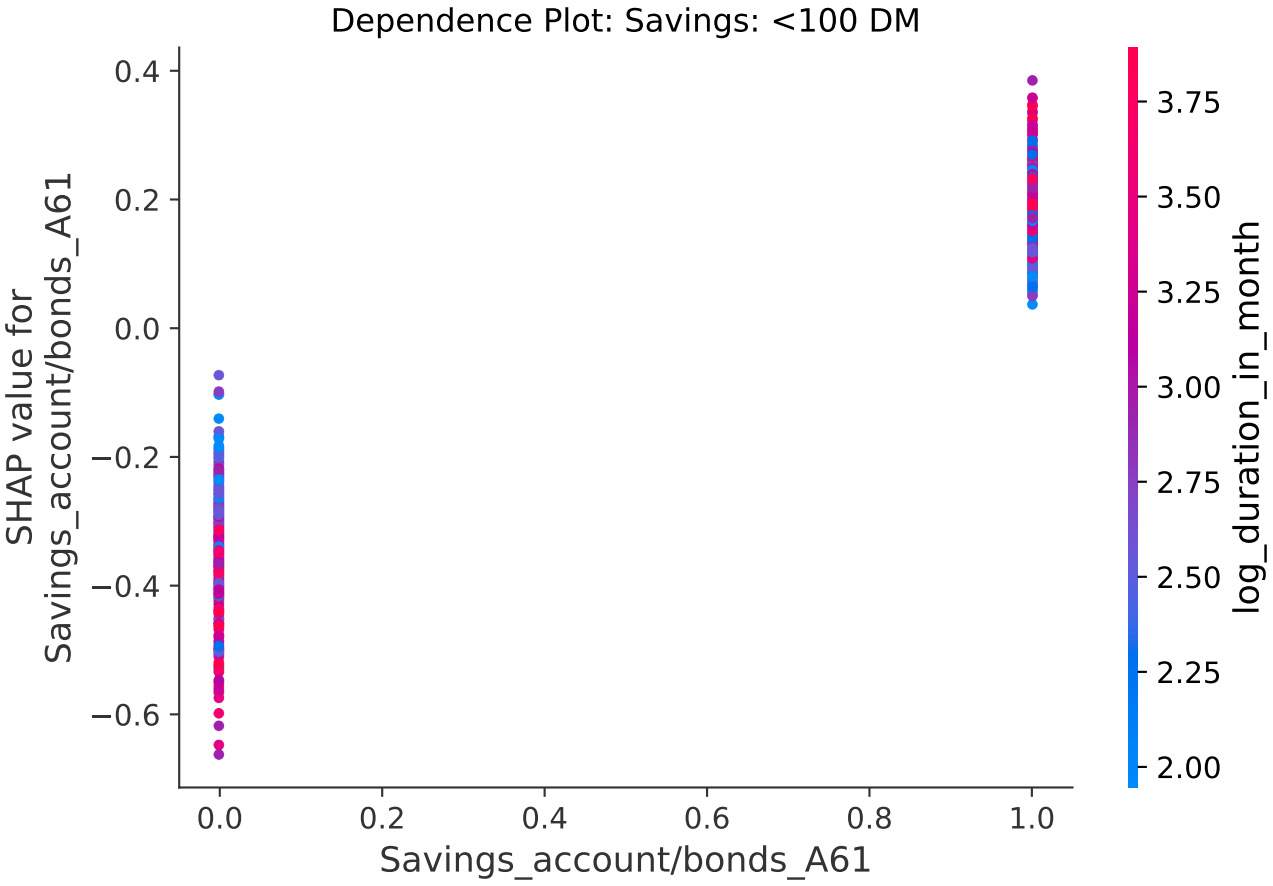

Dependence Plot: Credit-to-Age Ratio
SHAP value for credit_to_age
1.0
0.5
0.0
−0.5
−1.0
0
100
200
300
400
500
600
700
credit_to_age
Property_A121
1.0
0.5
0.0

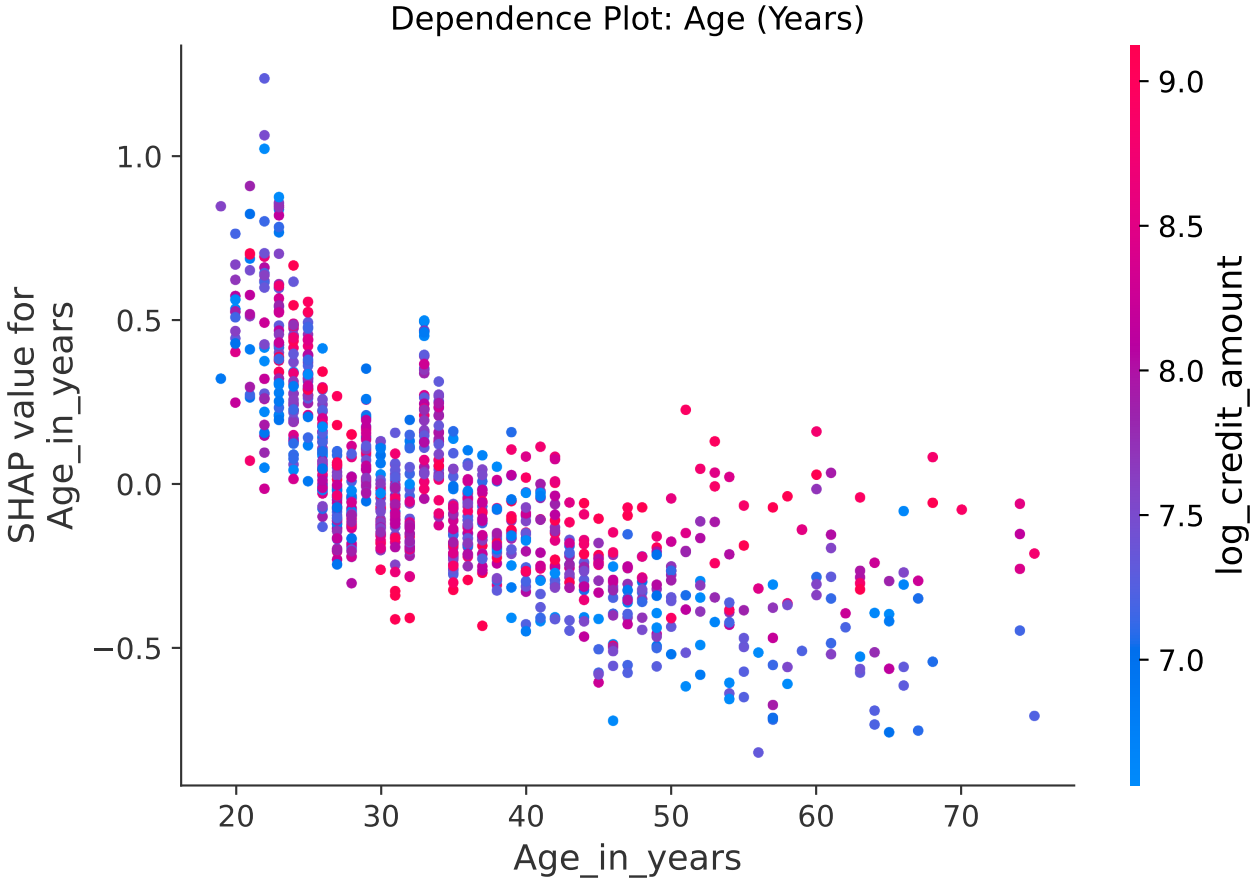

Dependence Plot: Duration (Months)
SHAP value for
Duration_in_month
1.0
0.5
0.0
−0.5
−1.0
−1.5
10
20
30
40
50
60
70
Duration_in_month
Property_A121
1.0
0.5
0.0

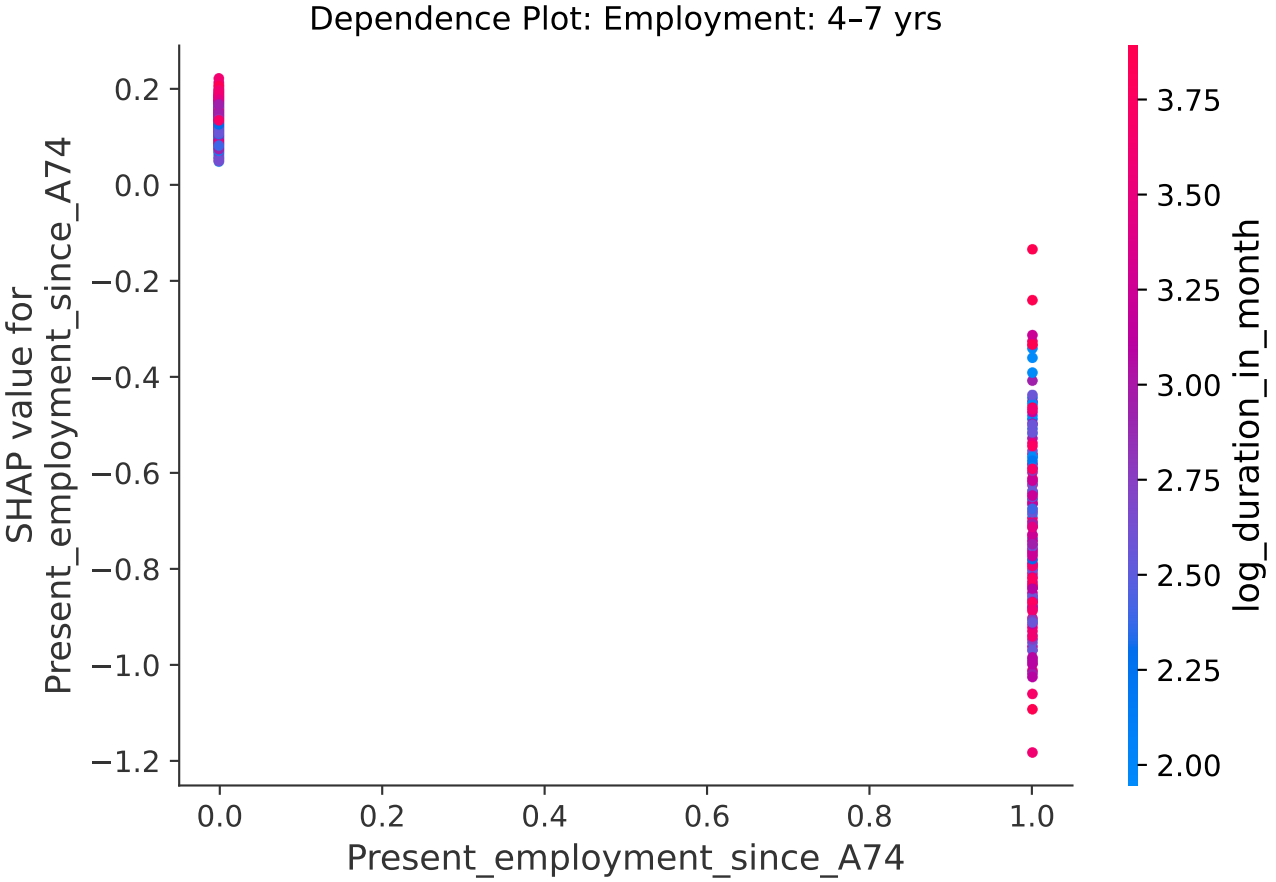

Dependence Plot: History: Critical/Other Loans
SHAP value for
Credit_history_A34
0.2
0.0
−0.2
−0.4
−0.6
−0.8
0.0
0.2
0.4
0.6
0.8
1.0
Credit_history_A34
Savings_account/bonds_A61
1.0
0.5
0.0

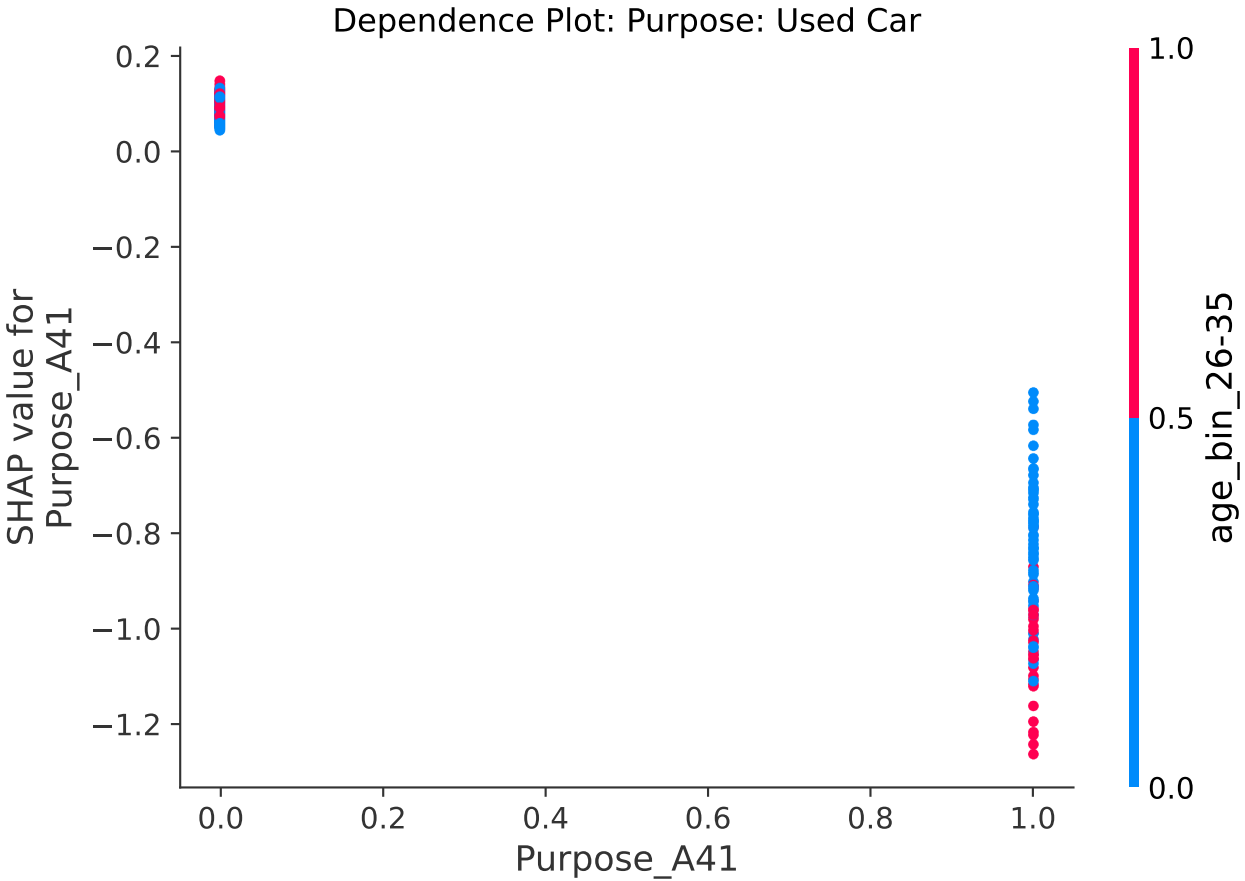

Dependence Plot: Checking: <0 DM
SHAP value for
Status_of_existing_checking_account_A11
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
0.0
0.2
0.4
0.6
0.8
1.0
Status_of_existing_checking_account_A11
1.0
0.5
0.0
Job_A173

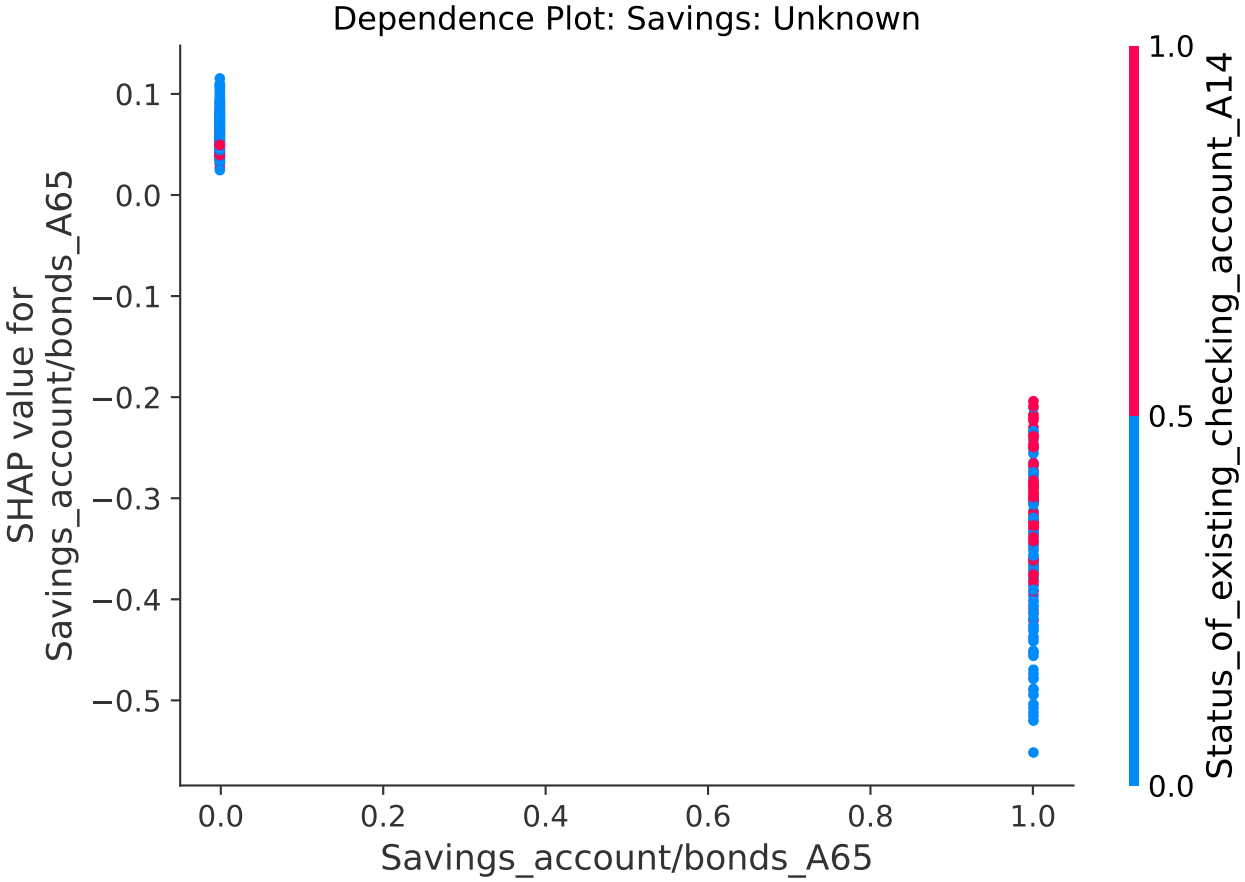

Dependence Plot: Purpose: New Car
SHAP value for Purpose_A40
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Purpose_A40
credit_per_month
350
300
250
200
150
100

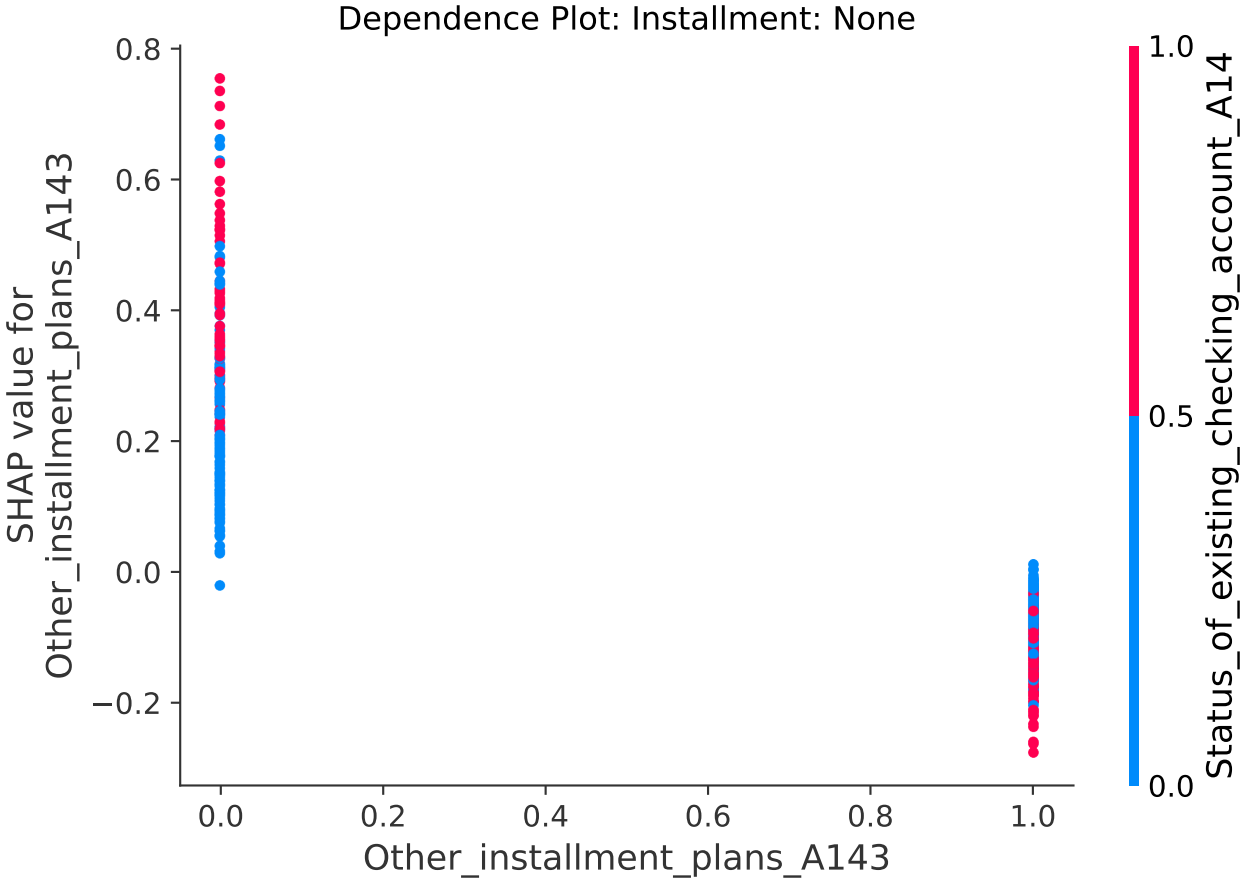

Dependence Plot: Savings: <100 DM
SHAP value for
Savings_account/bonds_A61
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.0
0.2
0.4
0.6
0.8
1.0
Savings_account/bonds_A61
log_duration_in_month
3.75
3.50
3.25
3.00
2.75
2.50
2.25
2.00

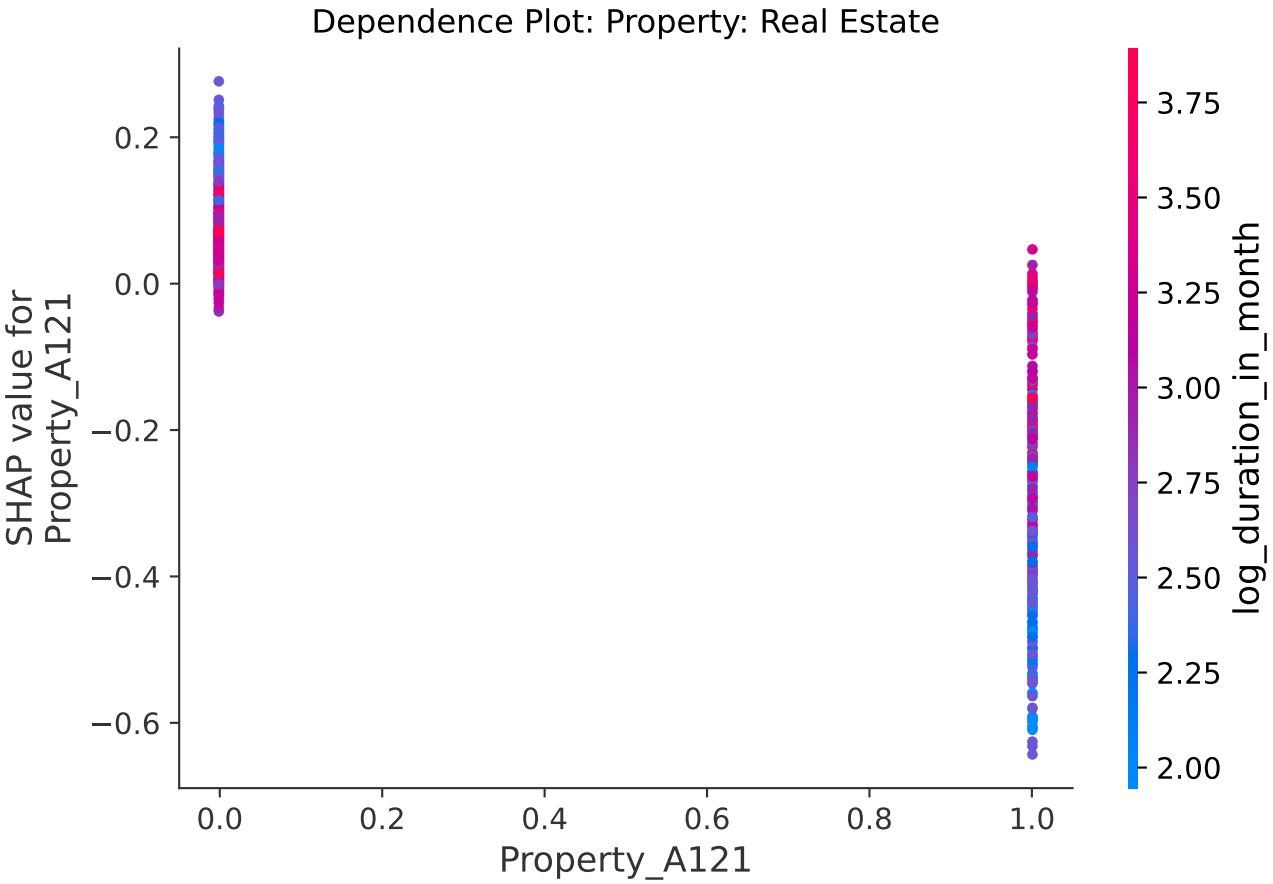

Dependence Plot: Age (Years)
SHAP value for
Age_in_years
1.0
0.5
0.0
−0.5
20
30
40
50
60
70
Age_in_years
9.0
8.5
8.0
7.5
7.0
log_credit_amount

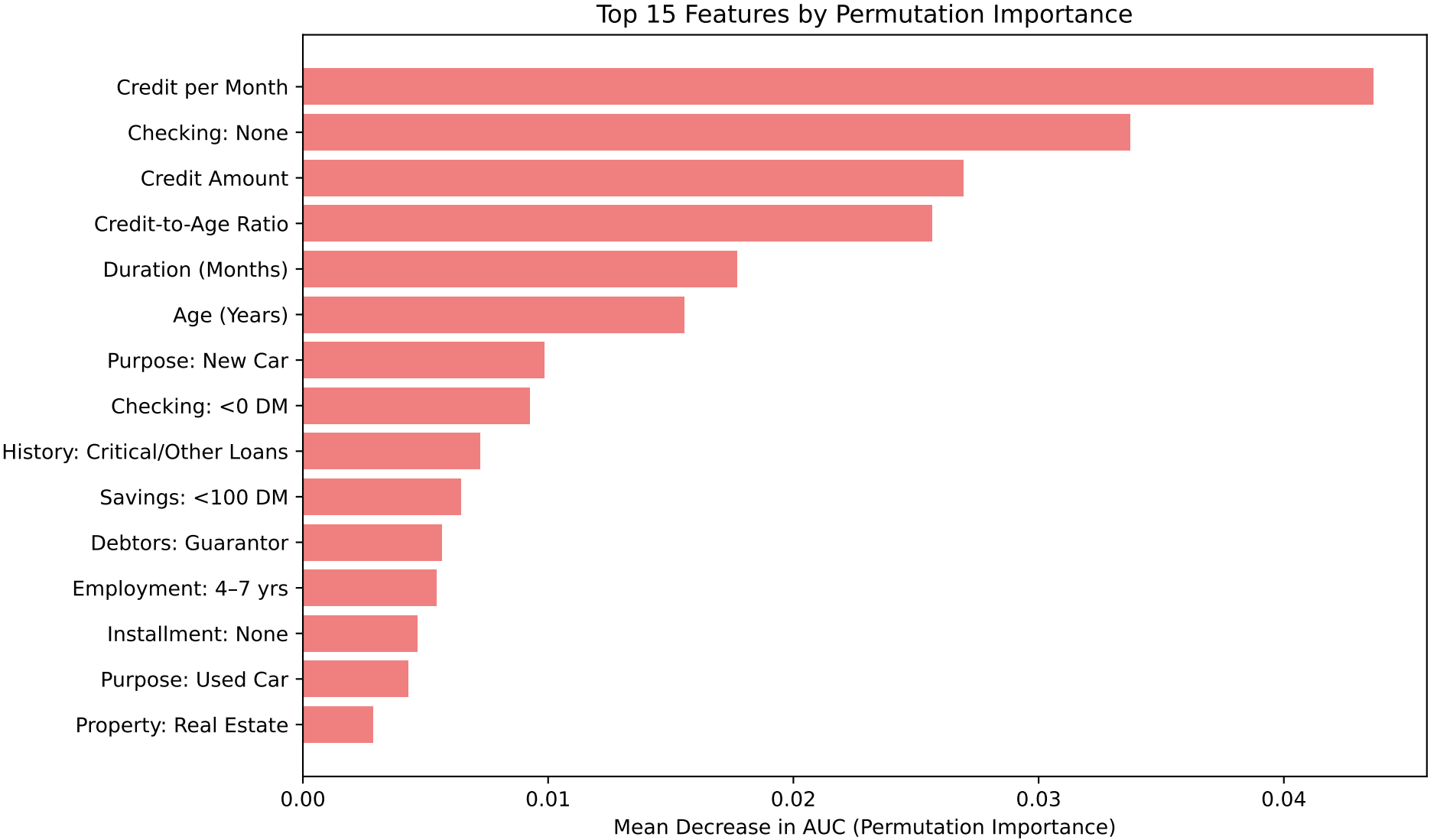

Dependence Plot: Employment: 4–7 yrs
SHAP value for
Present_employment_since_A74
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.2
0.4
0.6
0.8
1.0
Present_employment_since_A74
log_duration_in_month
3.75
3.50
3.25
3.00
2.75
2.50
2.25
2.00


Dependence Plot: Purpose: Used Car
SHAP value for Purpose_A41
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.2
0.4
0.6
0.8
1.0
Purpose_A41
1.0
0.5
0.0
age_bin_26-35


Dependence Plot: Savings: Unknown
SHAP value for
Savings_account/bonds_A65
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
0.0
0.2
0.4
0.6
0.8
1.0
Savings_account/bonds_A65
1.0
0.5
0.0
Status_of_existing_checking_account_A14


Dependence Plot: Installment: None
SHAP value for
Other_installment_plans_A143
0.8
0.6
0.4
0.2
0.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Other_installment_plans_A143
1.0
0.5
0.0
Status_of_existing_checking_account_A14


Dependence Plot: Property: Real Estate
SHAP value for Property_A121
0.2
0.0
−0.2
−0.4
−0.6
0.0
0.2
0.4
0.6
0.8
1.0
Property_A121
log_duration_in_month
3.75
3.50
3.25
3.00
2.75
2.50
2.25
2.00


Top 15 Features by Permutation Importance
Credit per Month
Checking: None
Credit Amount
Credit-to-Age Ratio
Duration (Months)
Age (Years)
Purpose: New Car
Checking: <0 DM
History: Critical/Other Loans
Savings: <100 DM
Debtors: Guarantor
Employment: 4–7 yrs
Installment: None
Purpose: Used Car
Property: Real Estate
0.00
0.01
0.02
0.03
0.04
Mean Decrease in AUC (Permutation Importance)